MUTED DISTILLATION: SEALING REASONING FROM TEACHER UNLOCKS TOKEN-EFFICIENT AGENTS

ABSTRACT

On-policy Distillation (OPD) is an important component for large language model post-training. However, its performance is reported to be negatively affected by the reasoning mismatch between the student and the teacher models. In this paper, we further illustrate that in multi-turn agentic tasks, such a discrepancy can also cause the student model to reason more inefficiently. After OPD training, the student's per-turn thinking tokens can grow by 2×, because the teacher keeps extending the student's thoughts rather than letting it conclude. We propose Muted Distillation (MD), which seals (or "mutes") the student's reasoning content from the teacher model's context, and find that training with it yields a far more token-efficient student with comparable or higher performance. For instance, on knowledge work benchmark OfficeQA Pro, MD improves pass@1 by 2.7 points over standard OPD while reducing output tokens per episode by 57%.

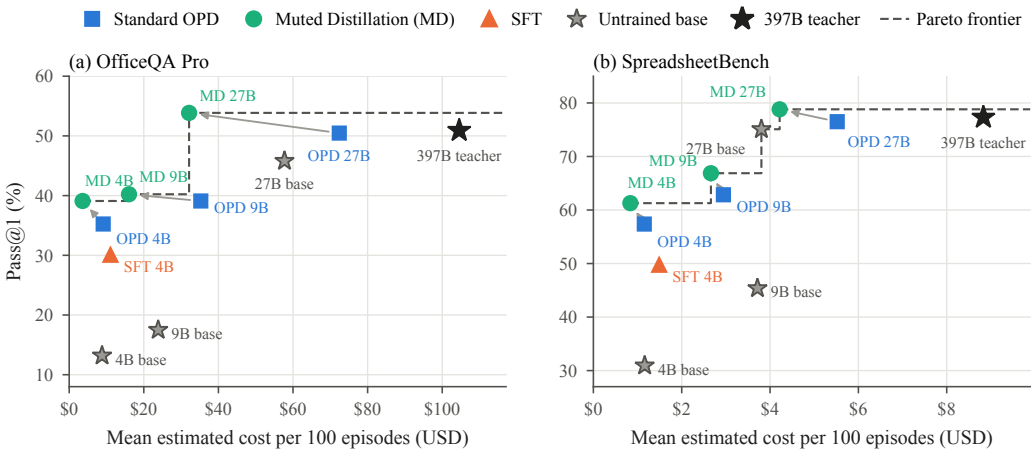


Figure 1: Pass@1 against the mean estimated inference cost per 100 episodes for Qwen3.5 students of 4B, 9B, and 27B parameters, each trained recipe at its best evaluated checkpoint. Cost counts uncached prompt and output tokens at the median verified paid-provider rate for each backbone (rates frozen on 2026-09-10).

1 INTRODUCTION

On-policy Distillation (OPD) has become a common post-training algorithm of large language models (Lu & Lab, 2025). In OPD, the student samples its own trajectories and the teacher scores every token, providing a dense, per-token learning signal that avoids both the exposure bias of supervised fine-tuning and the sparse rewards of reinforcement learning (Agarwal et al., 2024). Frontier models rely on OPD to consolidate domain-specialized teachers into a single deployable agent (Xiao et al., 2026; Xu et al., 2026; Zeng et al., 2026; Team et al., 2026).

However, because the teacher scores the student's tokens conditioned on the student's own reasoning, OPD is sensitive to the reasoning mismatch between the two models (Li et al., 2026). The

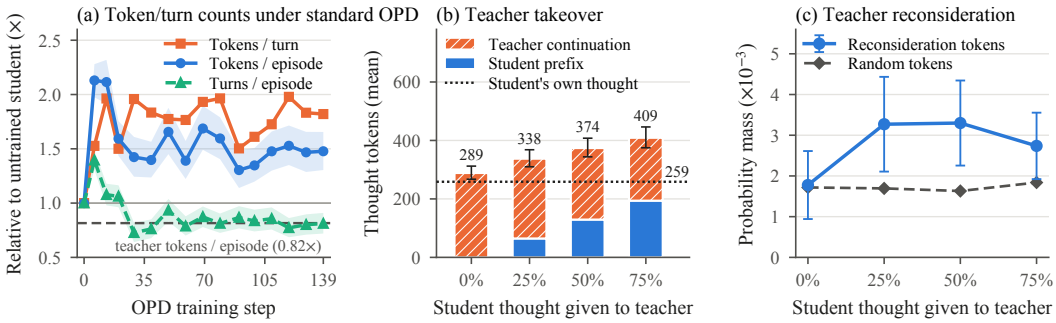


Figure 2: **OPD lengthens the student's reasoning, and reading the student's thought changes how the teacher continues it.** (a) A Qwen3.5-4B student trained with standard OPD from a Qwen3.5-397B-A17B teacher on OfficeQA Pro. Output tokens per episode, turns per episode, and output tokens per turn are shown relative to the student's base model. The dashed line marks the teacher's own tokens per episode. (b) The teacher continues the student's turn after reading 0–75% of the student's thought. Each bar stacks the mean student prefix and the mean teacher continuation, so its height is the mean total thought. The dotted line marks the mean length of the student's own thought. (c) The teacher's next-token probability mass on reconsideration tokens (e.g., "wait", "hmm", "actually") over its 24-token continuation, compared with a random token set of the same size and the same probability mass at the start.

sampled trajectories are on-policy for the student but "off-policy" for the teacher (Huang et al., 2026), which makes the teacher model drift from its own training distribution after seeing student's partial solutions (Ma et al., 2026). This can fundamentally shape the teacher's training signal because even a strong teacher that scores higher on benchmarks can fail to improve the student after OPD (Li et al., 2026). So far, however, this mismatch has been studied almost exclusively through the lens of task accuracy, leaving the discussion of its effects on token efficiency untouched.

Our study focuses on the multi-turn knowledge-work agents (Opsahl-Ong et al., 2026; Patwardhan et al., 2026; Ma et al., 2024) where token efficiency is especially crucial because the end-to-end latency matters for the user experiences (LangChain, 2026). We start by observing that OPD tends to increase the number of tokens generated per episode, even though the teacher is more token-efficient than the student (Figure 2a). Further analysis on the trajectories shows that the growth comes from the excessive thinking per turn rather than longer episodes (Figure 2a). To track where this extra reasoning comes from, we provide the teacher with a partial student thought and let it continue. We found that the more of student's thoughts the teacher reads, the longer the resulting reasoning becomes (Figure 2b). The decomposition of teacher continuations shows that the teacher doubles the probability of reconsideration tokens such as "wait", "hmm", "actually", while the baseline random tokens are unaffected (Figure 2c). This leads us to the hypothesis that the teacher is uncertain about the student's previous thoughts, and tries to extend the reasoning.

As a simple fix, we mask the thinking tokens when we calculate the policy loss, so that the teacher model only provides training signals on the tail tokens. This strategy yields a partial yemedy. On OfficeQA Pro, it reduces output tokens per episode by 26% relative to standard OPD, yet the student still generates 10% more tokens than before training, with a slight reduction on pass@1. This motivates our complete algorithm, Muted Distillation (MD), which goes one step further and removes the student's reasoning content from the teacher's view altogether. For each turn, the teacher scores the student's tail tokens, conditioned only on the interaction history and an empty thinking block, while the student is still trained on the same tokens in its own full context. The teacher thus judges what the student proposes *without* reading how it thought, and can no longer push the student to extend reasoning it did not write. Such a simple change results in a student both more accurate and far more efficient. Across two benchmarks, Muted Distillation matches or improves the accuracy of standard OPD while reducing output tokens per episode by up to 57%.

Overall, our contributions are:

- We delve into the extended reasoning after OPD training, and we identify the teacher-student reasoning inconsistency as a reason for the extra reasoning tokens generated by the student.
- We provide a comprehensive analysis by observing the student's generated tokens along with the effects of student prefix on teacher continuations.
- We evaluate the proposed Muted Distillation extensively on knowledge work agents, and proves that it can significantly reduce the number of generated tokens.

2 On-policy Distillation (OPD)

Suppose that we distill from the teacher model π_T to the student π_S with OPD. It first samples response y from π_S for task input x. Then it tries to align the student's probability distribution with teacher's using reverse Kullback-Leibler (KL)

$$\mathcal{L} = \mathbb{E}_{y\sim\pi_S}\Big[\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} D_{\mathrm{KL}}\big(\pi_S(\cdot \mid x, y_{<t})\big\|\pi_T(\cdot \mid x, y_{<t})\big)\Big] \tag{1}$$

The derivative of reverse KL can be rewritten into the same format as reinforcement learning,

$$\nabla_S\mathcal{L} = -\mathbb{E}_{y\sim\pi_S}\Big[\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} A_t\nabla_S \log(\pi_S(y_t \mid x, y_{<t}))\Big] \tag{2}$$

where $A_t = \log\pi_T(y_t \mid x, y_{<t}) - \log\pi_S(y_t \mid x, y_{<t})$ is the advantage. It represents how much more likely the teacher will generate the same token under the same prefix as the student. From the form of advantage, we can also see that reverse KL is "mode-seeking", because the student is penalized for placing mass where the teacher does not, but not for dropping modes the teacher covers (Agarwal et al., 2024). In practice, calculating the log difference between the entire probability distribution triggers out-of-memory issue during backward, so we only keep the log-prob difference for the one sampled token as suggested in Lu & Lab (2025).

The advantage of OPD over supervised fine-tuning (SFT) comes from the on-policy sampling process that prevents the exposure bias during test. On long-horizon agentic tasks, such an advantage can be especially enlarged due to the composed errors over turns. As show in Ross et al. (2011), such an error is quadratic with respect to the turns and linear with the imitation error during behavioral cloning.

3 Methodology

Motivation The motivation of this research is the observation that standard OPD often extends the token spent on each episode even though the teacher model is much more efficient (Figure 2a). Further decomposition shows that the per-turn number of tokens increases during training while the number of turns decreases. Upon categorizing the thought tokens based on the teacher–student agreement, in Figure 4b, we can see that the number of padding tokens increases significantly while the informative tokens stay at a similar amount. Thus the student learns to emit tokens that the teacher generally agree with. We then conduct a continuation experiment in Figure 2b, we let the teacher finish the student's thoughts. As the proportion of student's thoughts involved increases, we observe a clear trend of more thought tokens in total. When we look locally on what the teacher encourages, we discover that the teacher spends more reconsideration tokens during the continuations after viewing more of student's thoughts. And because of the mode-seeking behavior of reverse KL, the student sharpens onto the teachcer's preferred continuations and learns to keep reasoning instead of concluding early. Consequently, we arrive at the hypothesis that the extended thought comes from the teacher–student discrepancies which causes the teacher to be confused around the student's generations. We can also see this from Figure 4a where we show the values of KL around the `</think>` token, and we clearly observe a degradation on the teacher's criticism for tail tokens.

Muted Distillation In order to prevent the teacher from being distracted by the student's thinking process, we propose Muted Distillation, that removes the student's thought from the teacher's context and only calculate the advantage term on top of tail tokens. Formally, we write the student's response in a turn as $y = (z, a)$, where z is the thought enclosed in the thinking block and a is the

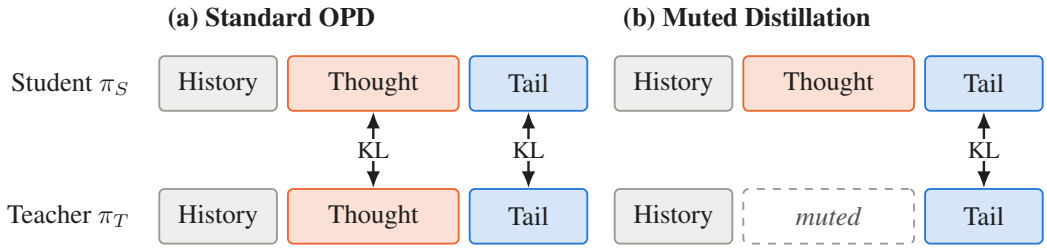


Figure 3: In each turn, standard OPD lets the teacher read the student's thought and applies the KL loss to both the thought and the tail. Muted Distillation replaces the thought with an empty block in the teacher's context and applies the loss to the tail only.

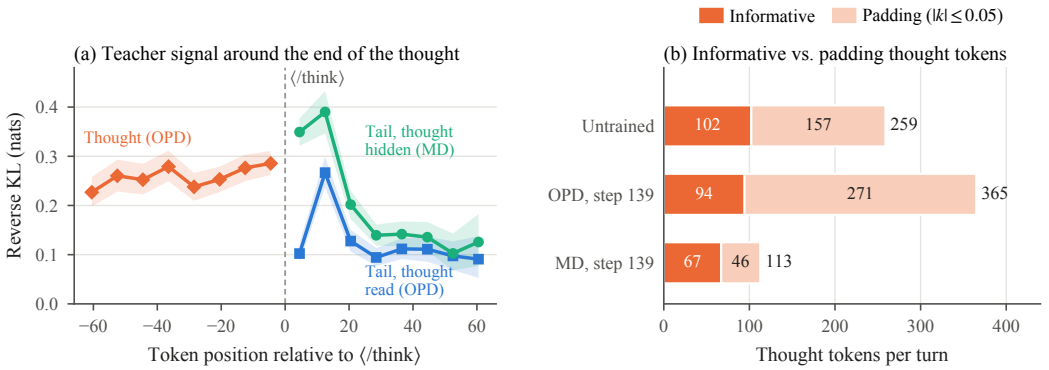


Figure 4: (a) Per-token reverse KL between the untrained Qwen3.5-4B student and the Qwen3.5-397B-A17B teacher around the end of the thought. The two tail curves score the same tokens with the teacher reading the thought (standard OPD) or with the thought hidden (Muted Distillation). (b) Thought tokens per turn, split into padding tokens on which the teacher already agrees with the student ($|k| \leq 0.05$) and informative tokens, all scored with the teacher reading the whole turn. Turns from OfficeQA Pro training prompts.

tail, i.e., the visible tokens after `</think>`. Let $\mathcal{Z}$ and $\mathcal{A}$ denote the positions of the thought and tail tokens in y, so that $\mathcal{T} = \mathcal{Z} \cup \mathcal{A}$, and let $\varnothing$ denote an empty thinking block. Muted Distillation optimizes

$$\mathcal{L}_{\text{MD}} = \mathbb{E}_{y \sim \pi_S}\Big[\frac{1}{|\mathcal{A}|}\sum_{t \in \mathcal{A}} D_{\text{KL}}\big(\pi_S(\cdot \mid x, y_{<t}) \big\| \pi_T(\cdot \mid x, \varnothing, a_{<t})\big)\Big], \tag{3}$$

where $a_{<t}$ denotes the tail tokens preceding position t. Compared with Eq. 1, the summation runs over the tail only, and the teacher's context replaces the student's thought z with $\varnothing$ while keeping the interaction history x and the visible tail. The student's own context is unchanged, so it still conditions on its full thought. The corresponding per-token advantage becomes

$$A_t = \begin{cases} \log \pi_T(y_t \mid x, \varnothing, a_{<t}) - \log \pi_S(y_t \mid x, y_{<t}) & t \in \mathcal{A}, \\ 0 & t \in \mathcal{Z}, \end{cases} \tag{4}$$

that is, the teacher judges each tail token given what the student has done so far, not how it reasoned, and the thought receives no direct gradient.

4 Results

4.1 Dataset

We focus on knowledge work agents in this paper because of its need of low latency during deployment, using OfficeQA Pro (Opsahl-Ong et al., 2026) and SpreadsheetBench (Ma et al., 2024).

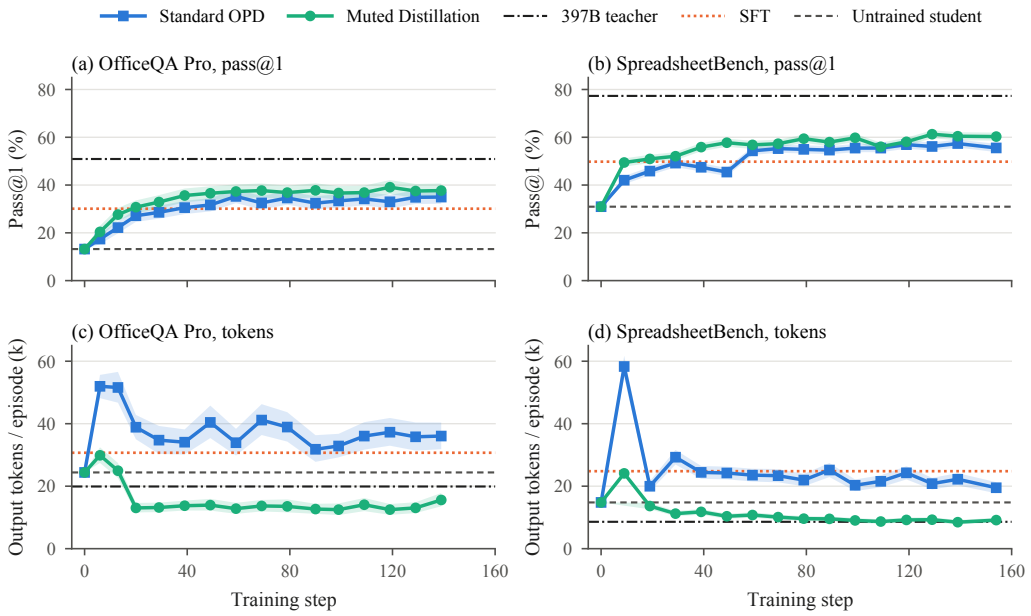


Figure 5: Offline evaluation of each checkpoint of a Qwen3.5-4B student trained with a Qwen3.5-397B-A17B teacher, on OfficeQA Pro and SpreadsheetBench Verified-400. Horizontal lines mark the teacher, an SFT baseline trained online on teacher demonstrations, and the untrained student, all evaluated under the same protocol.

OfficeQA Pro is an enterprise question-answering benchmark over U.S. Treasury Bulletins, a corpus of roughly 89,000 pages. Each question requires locating evidence in one or more bulletins. The benchmark has two splits, the 133 Pro questions and 113 easy questions. We train on the easy split only and report all results on the held-out Pro split. The agent works in a sandboxed container. Episodes are capped at 120 turns and 8,192 output tokens per turn.

SpreadsheetBench contains 912 spreadsheet manipulation instructions collected from Excel help forums, each paired with about three workbook test cases in which the data differ. A solution is correct on a test case if the designated answer cells of the produced workbook match the golden workbook, and an instruction counts as solved under the hard restriction only if all of the test cases pass. We train on 502 instructions that are disjoint from the 400 instructions verified from experts. The agent sees only the first test case's workbook and must write a reusable solution in its sandbox. The program is then replayed on every test case in a fresh, and the answer cells are compared with the golden workbook. Episodes are capped at 50 turns, and we report pass@1 under the hard restriction.

On both benchmarks, we sample 8 attempts per task at temperature 1.0 and report pass@1, and the mean number of output tokens per episode. Also, we choose mini-swe-agent as our harness for its simplicity and wide effectiveness (Yang et al., 2024).

4.2 Experimental Setup

We choose Qwen3.5-397B-A17B as the teacher model and Qwen3.5-4B, 9B, and 27B as the student models (Team, 2026). The teacher model is served with SGLang (Zheng et al., 2024) and used only to score the student's sampled tokens. Our training infrastructure is based on Slime (Zhu et al., 2025). All training uses 64k-token training context and at most 8,192 tokens per turn. Unless stated otherwise, all methods use Adam with a constant leanring rate of 10^{-6}, weight decay 0.1, sampling temperature 1.0, and one optimizer step per rollout batch, with the per-token loss averaged over all supervised tokens in the batch.

For OPD and Muted Distillation, each training step samples 16 prompts. We train for 139 steps on OfficeQA Pro and 154 steps on SpreadsheetBench. The 9B and 27B students use the same recipe. Dr. GRPO is a pure reinforcement learning algorithm we trained as a comparison (Liu et al., 2025),

Table 1: Pass@1 (%) and output tokens per episode (k) of the Qwen3.5-4B student at the final checkpoint (OfficeQA Pro: step 139; SpreadsheetBench: step 154), with the same evaluation protocol as Figure 5. Dr. GRPO is trained with the task reward instead of a teacher and is shown as a reinforcement-learning reference. Bold marks the best distilled student in each column.

			OfficeQA Pro		SpreadsheetBench	
Method	Teacher sees thought	Loss on	Pass@1	Tokens	Pass@1	Tokens
Untrained student	–	–	13.2	24.4	30.9	14.8
SFT	–	–	30.1	30.7	49.8	24.8
Dr. GRPO (RL)	–	–	42.4	13.3	57.7	35.3
Teacher (397B)	–	–	50.9	19.9	77.3	8.6
Standard OPD	✓	whole response	35.0	36.0	55.5	19.5
Mask-think-tail	✓	tail	32.0	26.8	55.5	18.3
Think-only	✓	thought	30.9	27.4	57.9	16.7
Muted Distillation	×	tail	**37.7**	**15.6**	**60.3**	**9.1**

where group mean is centered without standard deviation and a fixed length denominator is applied equal to the 8,192 budget. The reward is the exact match on OfficeQA Pro and the soft-restriction score on SpreadsheetBench. We use a group size of 8. SFT is an unsupervised baseline, where we use the teacher to generate expert off-policy trajectories online and distill onto the student.

4.3 FINDINGS

MD matches OPD's accuracy at far fewer tokens. Figure 5 tracks both students over training and Table 1 reports the final checkpoints. On accuracy, MD is never behind standard OPD at any evaluated checkpoint. It ends at 37.7 vs. 35.0 pass@1 on OfficeQA Pro and 60.3 vs. 55.5 on SpreadsheetBench, and it reaches OPD's final accuracy early, within 40 steps on OfficeQA Pro and 60 steps on SpreadsheetBench. On tokens, the two recipes diverge from the first evaluated checkpoint and never cross. Standard OPD inflates the student, and it peaks at 52k tokens per episode on OfficeQA Pro and stays at 32–41k for the rest of training, 48% above the untrained student's 24.4k, while MD drops to 13k by step 20 and stays there, ending at 15.6k, 57% below OPD and 36% below the untrained student. SpreadsheetBench shows the same pattern. OPD settles at 20–25k tokens per episode, above the untrained student's 14.8k, whereas MD falls below 10k and ends at 9.1k, 53% below OPD. MD is the only distilled student that is more token-efficient than the untrained one, and it approaches the teacher's own budget (19.9k and 8.6k), while standard OPD moves away from it despite distilling from the same teacher. The SFT baseline trained on the teacher's demonstrations is both less accurate (30.1 and 49.8) and longer than the untrained student (30.7k and 24.8k), so imitating the teacher's traces does not transfer its efficiency either. Finally, MD also outperforms Dr. GRPO on SpreadsheetBench that is trained with golden verifiers, showing the generalizability of MD.

MD's improvements come from not just loss masking. MD differs from standard OPD in two ways: the loss is restricted to the tail, and the teacher no longer reads the thought. Table 1 separates the two. Mask-think-tail keeps the loss on the same tail tokens as MD but still lets the teacher read the thought, so the gap between these two rows isolates muting. Restricting the loss alone is only a partial remedy because it lowers tokens per episode by 26% on OfficeQA Pro (36.0k to 26.8k) and by 6% on SpreadsheetBench (19.5k to 18.3k), leaving the student 10% and 24% above its untrained length, and it costs 3.0 points of pass@1 on OfficeQA Pro. Hiding the thought as well removes the remaining growth and recovers the accuracy. MD ends 42% below mask-think-tail on OfficeQA Pro and 50% below it on SpreadsheetBench, with 5.7 and 4.8 more points of pass@1. Supervising the thought alone does not help either. Think-only trims tokens by 24% and 14% relative to OPD, but the student still thinks longer than before training, and its OfficeQA Pro accuracy falls to 30.9. This ordering is what the analysis in Section 3 predicts. When the teacher reads the thought, its correction on the tail is weak, so a tail-only loss with a reading teacher learns from a diluted signal. Muting therefore does two things that masking cannot: it strengthens the signal on the tokens that are trained, and it stops the teacher from encouraging the tokens that are not.

Table 2: Transfer of the same final checkpoints as Table 1 to the harder OfficeQA Pro v2 and SpreadsheetBench 2. Score is pass@1 (%) on OfficeQA Pro v2 and target-cell accuracy (%) on SpreadsheetBench 2. Bold marks the best trained student in each column.

	OfficeQA Pro v2			SpreadsheetBench 2		
Method	Score	Tokens	Cost	Score	Tokens	Cost
Untrained student	8.2	30.8	\$11.8	26.4	25.8	\$4.1
Standard OPD	10.8	53.1	\$18.2	34.8	29.1	**\$4.4**
Muted Distillation	**15.1**	**31.1**	**\$11.0**	**37.6**	**26.0**	\$4.7

MD's advantage holds across student sizes and translates into cost. Figure 1 places each recipe's best checkpoint on an accuracy–cost plane for 4B, 9B, and 27B students, where cost prices the prompt and output tokens of an episode at public API rates. At every size, MD is at least as accurate as standard OPD and cheaper to run. On OfficeQA Pro it cuts the cost per 100 episodes by 61% at 4B (\$3.6 vs. \$9.1), 55% at 9B, and 56% at 27B, while adding 3.9, 1.1, and 3.4 points of pass@1; on SpreadsheetBench the savings are 27%, 10%, and 24% with gains of 3.9, 4.1, and 2.3 points. The MD students therefore define the Pareto frontier on both benchmarks. Two consequences stand out. First, muting moves a student down a size class: the 4B MD student reaches the same OfficeQA Pro accuracy as the 9B OPD student (39.1 vs. 39.1) at a tenth of the cost (\$3.6 vs. \$35.3), and it is cheaper than the untrained 4B model itself (\$8.8) while almost tripling its accuracy. Second, the 27B MD student overtakes its own 397B teacher on both benchmarks (53.9 vs. 50.9 and 78.8 vs. 77.3) at 31% and 48% of the teacher's cost, whereas the 27B OPD student does not (50.5 and 76.5) and costs 1.3–1.6$\times$ more than its muted counterpart.

Gains transfer to harder benchmarks. Table 2 evaluates the same final checkpoints on two harder, held-out variants. OfficeQA Pro v2 is composed of new questions over a separate corpus, and SpreadsheetBench 2 consists of tasks that are end-to-end spreadsheet workflows rather than single instructions. Neither was used for training or for checkpoint selection. On OfficeQA Pro v2, standard OPD improves the untrained student by only 2.6 points while inflating its episodes by 72% (30.8k to 53.1k tokens); MD nearly doubles the untrained accuracy (8.2 to 15.1, 4.3 points above OPD) with episodes no longer than the untrained student's (31.1k), at 60% of OPD's cost. On SpreadsheetBench 2, MD again leads OPD in accuracy (37.6 vs. 34.8 target-cell accuracy) and uses 11% fewer output tokens, although its episodes contain slightly more prompt tokens, so the cost is marginally higher (\$4.7 vs. \$4.4 per 100 episodes). We report target-cell accuracy here because strict whole-task accuracy is below 2% for every 4B student on this benchmark; the two trained students are separated by 27 vs. 16 fully solved attempts out of 2,376, in the same direction. The accuracy advantage of MD thus survives a distribution shift in both the questions and the corpus, and the token saving survives it where the student's inflation under OPD is large.

MD raises student-teacher agreement. Finally, we ask whether the student actually ends up closer to the teacher on the tokens that matter for the task. Figure 6 measures, at every checkpoint, the overlap between the student's and the teacher's top-16 next-token candidates on the student's own tail tokens, with each student scored under the teacher context it was trained with. Before training the two contexts agree with the student about equally (57.6% with the thought visible, 56.3% with it hidden), so the hidden context does not start with an advantage. Under MD the overlap rises within the first evaluated checkpoint and stays high, ending at 64.5% on OfficeQA Pro and 63.5% on SpreadsheetBench, 8.2 and 6.3 points above its starting value. Under standard OPD it barely moves on OfficeQA Pro (58.4% at the end) and drifts downward on SpreadsheetBench (58.7% to 55.5%). The student's visible actions move no closer to what the teacher would do, even though the loss is minimized on every token of the response. This is the same asymmetry as in Figure 4a, seen from the student's side. With the thought hidden, the tail is where the teacher disagrees and where the gradient goes, and the student's tail converges toward the teacher. With the thought visible, the teacher's disagreement is spent on the thought, and the tail is left alone.

Thought shrinks after training with MD. Figure 7 shows a turn from each final checkpoint on the same OfficeQA Pro task at the same stage of the episode. The difference is not that the MD student stops reasoning. It still opens a thought, but the thought is short and moves directly to the next action, and the explanation that OPD buries in the thought appears instead as visible prose before

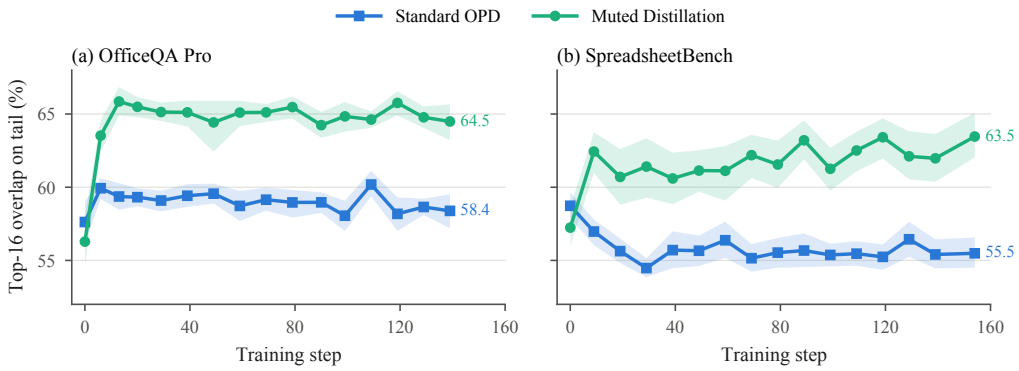


Figure 6: **Muted Distillation raises student–teacher agreements.** Top-16 candidate overlap between the Qwen3.5-4B student and the Qwen3.5-397B-A17B teacher on student's responses, i.e., the number of next-token candidates shared by both models' top-16 divided by 16, measured at each training checkpoint on that checkpoint's own trajectories.

the command, where the teacher could see and correct it. The OPD student restates the question at length inside its thought, which the muted teacher never endorsed, and then issues a broad search. This pattern holds for more data. Over all turns with a closed thought at the final OfficeQA Pro checkpoints, the thought shrinks from 540 to 98 tokens on average while visible prose grows from 6 to 267 tokens and commands lengthen slightly from 83 to 130, so MD relocates explanation rather than removing it.

5 Related Works

On-policy Distillation. On-policy distillation trains the student on its own samples with a per-token signal from the teacher, which avoids the exposure bias of imitating teacher-written traces and the sparsity of outcome rewards (Agarwal et al., 2024; Gu et al., 2024; Lu & Lab, 2025). It has become a standard stage in frontier post-training pipelines, where it consolidates specialized teachers into one deployable model (Xiao et al., 2026; Xu et al., 2026; Zeng et al., 2026; Team et al., 2026). A growing line of work studies what happens when the teacher is asked to score trajectories it did not write. The teacher's conditionals drift once it is conditioned on a student prefix (Li et al., 2026; Huang et al., 2026), its ability to teach decays along the sequence (Liu et al., 2026a), its stopping behavior can disagree with the student's and inflate responses (Yang et al., 2026a), and a stronger teacher does not reliably produce a better student (Li et al., 2026; Zhao & Yu, 2026). Proposed remedies mostly act on the teacher or on the target. Training the teacher on student-induced states before distillation (Ma et al., 2026), updating it jointly with the student (Huang et al., 2026), filtering or reweighting the per-token signal (Shen et al., 2026; Wang et al., 2026; Yang et al., 2026b), or warming the student up so that its prefixes are closer to the teacher's (Liu et al., 2026b). Several of these works also report that OPD lengthens responses (Yang et al., 2026a; Luo et al., 2026a), but they treat it as an instability to be stabilized rather than as a cost in deployment. Muted Distillation is complementary to all of them. It changes neither the teacher nor the loss weighting, only what the teacher is allowed to see, and it needs no additional training or data. The closest relatives are methods that drop reasoning from the distillation target so that the student learns to answer with less or no visible thinking (Kalwar et al., 2026; Kaur et al., 2026); Muted Distillation instead keeps the student's thought intact and in its own context, and removes it only from the teacher's.

Token-efficiency during Post-Training. Post-training with reinforcement learning tends to lengthen reasoning, and in agentic settings long reasoning can crowd out interaction with the environment (Cuadron et al., 2025). Part of the growth is an artifact of the objective. Length-normalized policy gradients reward longer incorrect responses, which Dr. GRPO removes with a constant normalizer (Liu et al., 2025). Most approaches to efficiency then add an explicit length objective: budget- or length-conditioned training (Aggarwal & Welleck, 2025; Muennighoff et al., 2025), length penalties and long-to-short distillation (Team et al., 2025; Luo et al., 2026b; Hou et al.,

Standard OPD, turn 6: 776 tokens

THOUGHT (723 tokens)

I see that the table at line 873–893 shows Trust account expenditures data. Looking at the columns, I need to understand what the question is asking for: 1. "investments expenditures" – This appears to be the "Investments in Federal securities > By trust accounts" column 2. "all other expenditures" – This would be everything else in Trust account expenditures [... 642 more tokens of thought omitted ...]

PROSE (0 tokens)

COMMAND (44 tokens)

```
grep -Rni "November\|December"
officeqa_corpus/
treasury_bulletin_1948_12.txt |
grep -i "trust\|expenditure" |
head -50
```

Muted Distillation, turn 6: 119 tokens

THOUGHT (36 tokens)

The search didn't find that exact table title. Let me try searching for similar patterns in the 1948 bulletins, particularly looking at different variations of table titles.

PROSE (43 tokens)

Let me find the table that contains the monthly data for trust account expenditures. I noticed line 550 has the column headers for investments and other expenditures. Let me look at the full context around that table.

COMMAND (30 tokens)

```
sed -n '540,600p' officeqa_corpus/
treasury_bulletin_1948_07.txt
```

Figure 7: One turn from each final Qwen3.5-4B checkpoint on the same OfficeQA Pro task, at the same stage of the episode. The standard OPD student spends 723 tokens re-reading the question inside its thought and writes no visible explanation; the Muted Distillation student writes a 36-token thought, states its plan in 43 tokens of visible prose, and issues a more targeted command.

2025), or adaptive policies that decide when to think at all (Zhang et al., 2025). These methods trade accuracy for length along a curve chosen by a hyperparameter, and they presuppose that the training signal itself is length-neutral. Our setting is different in that the extra tokens come from the distillation signal, which encourages the student's continued reasoning once the teacher has read it. Muted Distillation removes that source rather than penalizing its symptom, and it does so without a length term, so the student ends up shorter than before training while gaining accuracy. An explicit length objective could still be layered on top of it, which we leave to future work.

6 Conclusion and Discussion

On-policy distillation suffers from the reasoning discrepancies between teacher and student, while the empirical results on the issue mostly focuses on the accuracy. We have argued in this paper that such a mismatch can also cause the token costs to increase significantly on multi-turn knowledge work agents. Specifically, a Qwen3.5-4B student distilled from a teacher that is more token-efficient than itself ends up thinking roughly twice as long per turn. Our teacher-continuation experiment reveals that the teacher, once it reads the student's partial thought, prefers to extend that thought rather than conclude it. Then reverse KL's mode-seeking behavior encourages this preference. The extra tokens are mostly ones the teacher already agrees with, so nothing in the objective pushes back on them.

Muted Distillation addresses the issue by replacing the student's thought with an empty block in the teacher's context and gather loss only from the tail tokens, it removes the teacher's opportunity to extend the student's reasoning while leaving the student free to reason in its own context. The change is a few lines in the rollout code and adds no training, data, or hyperparameters. Across OfficeQA Pro and SpreadsheetBench, it matches or improves the accuracy of standard OPD while cutting output tokens per episode by more than half. It also moves 4B, 9B, and 27B students onto the accuracy–cost frontier and we prove that this performance transfers to harder held-out variants of both benchmarks, and brings the student's visible actions measurably closer to the teacher's.

In practice, teacher models are always imperfect, and there are some biases in their produced scores that are irrelevant with the desired capability. And following Goodhart's law, overoptimizing on such a proxy reward can result in drifting from the true objectives (Gao et al., 2023). And indeed this explains the numerous reports of empirical failures using the on-policy distillation objective (Fu et al., 2026; Zhu et al., 2026; Kaur et al., 2026). Our study sheds light on the potential of debiasing

the teacher model through the delibrate pruning of its contexts. While in this work we have proved that the teacher model can provide better scores without seeing student's reasoning, we want to point out that a more generic approach might be to selectively prune contexts based on the teacher's preference so that the sample is more "scorable". For instance, a naive approach would be to take the teacher's probability into consideration during student's sampling. In this way, the samples can potentially induce more accurate scores from the teacher. Due to the limitated resources, we leave this exploration to the future works.

REFERENCES

Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*, volume 2024, pp. 21246–21263, 2024.

Pranjal Aggarwal and Sean Welleck. L1: Controlling how long a reasoning model thinks with reinforcement learning. *arXiv preprint arXiv:2503.04697*, 2025.

Alejandro Cuadron, Dacheng Li, Wenjie Ma, Xingyao Wang, Yichuan Wang, Siyuan Zhuang, Shu Liu, Luis Gaspar Schroeder, Tian Xia, Huanzhi Mao, et al. The danger of overthinking: Examining the reasoning-action dilemma in agentic tasks. *arXiv preprint arXiv:2502.08235*, 2025.

Yuqian Fu, Haohuan Huang, Kaiwen Jiang, Jiacai Liu, Zhuo Jiang, Yuanheng Zhu, and Dongbin Zhao. Revisiting on-policy distillation: Empirical failure modes and simple fixes. *arXiv preprint arXiv:2603.25562*, 2026.

Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International conference on machine learning*, pp. 10835–10866. PMLR, 2023.

Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. Minillm: Knowledge distillation of large language models. In *International Conference on Learning Representations*, volume 2024, pp. 32694–32717, 2024.

Bairu Hou, Yang Zhang, Jiabao Ji, Yujian Liu, Kaizhi Qian, Jacob Andreas, and Shiyu Chang. Thinkprune: Pruning long chain-of-thought of llms via reinforcement learning. *arXiv preprint arXiv:2504.01296*, 2025.

Langlin Huang, Hao Liu, Mononito Goswami, Xinyu Li, Prithwith Jana, Nikos Kanakaris, Patrick Blöbaum, and Purak Jain. On the off-policy teacher in on-policy distillation. *arXiv preprint arXiv:2609.38360*, 2026.

Durgesh Kalwar, Vardhan Palod, and Subbarao Kambhampati. Masked distillation: Internalizing the chain-of-thought in language models. *arXiv preprint arXiv:2607.22629*, 2026.

Simran Kaur, Narutatsu Ri, Yinghui He, Liam Fowl, and Sanjeev Arora. Rethinking on-policy self-distillation for thinking models. *arXiv preprint arXiv:2607.05184*, 2026.

LangChain. State of agent engineering. `https://www.langchain.com/state-of-agent-engineering`, 2026. Accessed 2026-10-27.

Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huanang Gao, Wenkai Yang, Zhiyuan Liu, et al. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe. *arXiv preprint arXiv:2604.13016*, 2026.

Kaiyuan Liu, Ziyuan Zhuang, Yang Bai, Bing Wang, Rongxiang Weng, and Jieping Ye. Prefix teach, suffix fade: Local teachability collapse in strong-to-weak on-policy distillation. *arXiv preprint arXiv:2605.13643*, 2026a.

Tao Liu, Taiqiang Wu, Mao Zheng, Xuan Luo, Runming Yang, Xuewei Yang, Junjie Wang, and Yujiu Yang. Simple-opd: Demystifying warm-up for on-policy distillation. *arXiv preprint arXiv:2608.06802*, 2026b.

Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding r1-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.

Kevin Lu and Thinking Machines Lab. On-policy distillation. *Thinking Machines Lab: Connectionism*, 2025. doi: 10.64434/tml.20251026. https://thinkingmachines.ai/blog/on-policy-distillation.

Feng Luo, Yu-Neng Chuang, Guanchu Wang, Zicheng Xu, Xiaotian Han, Tianyi Zhang, and Vladimir Braverman. Demystifying opd: Length inflation and stabilization strategies for large language models. *arXiv preprint arXiv:2604.08527*, 2026a.

Haotian Luo, Haiying He, Yibo Wang, Shiwei Liu, Wei Li, Xiaochun Cao, Dacheng Tao, Naiqiang Tan, and Li Shen. O1-pruner: Length-harmonizing fine-tuning for o1-like reasoning pruning. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 14242–14257, 2026b.

Xiaoyu Ma, Haoyue Liu, Zhichao Wang, Jionghao Zhu, and Xiaoying Tang. How should teachers be prepared? rl on student-induced states for on-policy distillation. *arXiv preprint arXiv:2610.04950*, 2026.

Zeyao Ma, Bohan Zhang, Jing Zhang, Jifan Yu, Xiaokang Zhang, Xiaohan Zhang, Sijia Luo, Xi Wang, and Jie Tang. Spreadsheetbench: Towards challenging real world spreadsheet manipulation. *Advances in Neural Information Processing Systems*, 37:94871–94908, 2024.

Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori B Hashimoto. s1: Simple test-time scaling. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 20286–20332, 2025.

Krista Opsahl-Ong, Arnav Singhvi, Jasmine Collins, Ivan Zhou, Cindy Wang, Ashutosh Baheti, Owen Oertell, Jacob Portes, Sam Havens, Erich Elsen, et al. Officeqa pro: An enterprise benchmark for end-to-end grounded reasoning. *arXiv preprint arXiv:2603.08655*, 2026.

Tejal Patwardhan, Rachel Dias, Elizabeth Proehl, Grace Kim, Michele Wang, Olivia Watkins, Simon Fishman, Marwan Aljubeh, Phoebe Thacker, Laurance Fauconnet, et al. Gdpval: Evaluating ai model performance on real-world economically valuable tasks. In *International Conference on Learning Representations*, volume 2026, pp. 24005–24040, 2026.

Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pp. 627–635. JMLR Workshop and Conference Proceedings, 2011.

Zhanming Shen, Jintao Tong, Shaotian Yan, Chen Shen, Hao Chen, Wentao Ye, Xiaomeng Hu, Rui Miao, Haobo Wang, Junbo Zhao, et al. Purified opsd: On-policy self-distillation without losing how to think. *arXiv preprint arXiv:2607.02234*, 2026.

Kimi Team, Angang Du, Bofei Gao, Bowei Xing, Changjiu Jiang, Cheng Chen, Cheng Li, Chenjun Xiao, Chenzhuang Du, Chonghua Liao, et al. Kimi k1. 5: Scaling reinforcement learning with llms. *arXiv preprint arXiv:2501.12599*, 2025.

Kimi Team, Tongtong Bai, Yifan Bai, Yiping Bao, Jianfeng Cai, Xinyuan Cai, Peizhou Cao, Yuxuan Cao, Ziwei Chai, Y Charles, et al. Kimi k3: Open frontier intelligence. *arXiv preprint arXiv:2607.24653*, 2026.

Qwen Team. Qwen3.5: Towards native multimodal agents, February 2026. URL `https://qwen.ai/blog?id=qwen3.5`.

Yuanyi Wang, Su Lu, Yanggan Gu, Pengkai Wang, Yifan Yang, Zhaoyi Yan, Congkai Xie, Jianmin Wu, and Hongxia Yang. Not all disagreement is learnable: Token teachability in on-policy distillation. *arXiv preprint arXiv:2605.26844*, 2026.

Bangjun Xiao, Bingquan Xia, Bo Yang, Bofei Gao, Bowen Shen, Chen Zhang, Chenhong He, Chiheng Lou, Fuli Luo, Gang Wang, et al. Mimo-v2-flash technical report. *arXiv preprint arXiv:2601.02780*, 2026.

Anyi Xu, Bangcai Lin, Bing Xue, Bingxuan Wang, Bingzheng Xu, Bochao Wu, Bowei Zhang, Chaofan Lin, Chen Dong, Chenchen Ling, et al. Deepseek-v4: Towards highly efficient million-token context intelligence. *arXiv preprint arXiv:2606.19348*, 2026.

John Yang, Carlos E Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik R Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL https://arxiv.org/abs/2405.15793.

Yuxiao Yang, Tianrun Yu, Shangzhe Li, Kaixiang Zhao, Xuchao Zhang, Chetan Bansal, Huaxiu Yao, Taylor W Killian, and Weitong Zhang. When eos tokens disagree: Understanding length inflation in on-policy distillation. *arXiv preprint arXiv:2609.20511*, 2026a.

Zhicheng Yang, Zhijiang Guo, Yifan Song, Minrui Xu, Yongxin Wang, Yiwei Wang, Xiaodan Liang, and Jing Tang. Prune-opd: Efficient and reliable on-policy distillation for long-horizon reasoning. *arXiv preprint arXiv:2605.07804*, 2026b.

Aohan Zeng, Xin Lv, Zhenyu Hou, Zhengxiao Du, Qinkai Zheng, Bin Chen, Da Yin, Chendi Ge, Chenghua Huang, Chengxing Xie, et al. Glm-5: from vibe coding to agentic engineering. *arXiv preprint arXiv:2602.15763*, 2026.

Jiajie Zhang, Nianyi Lin, Lei Hou, Ling Feng, and Juanzi Li. Adaptthink: Reasoning models can learn when to think. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 3716–3730, 2025.

Shu Zhao and Tan Yu. Capable teacher, myopic scorer: Reward-free teacher repair for agentic on-policy distillation. *arXiv preprint arXiv:2610.28586*, 2026.

Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody H Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E Gonzalez, et al. Sglang: Efficient execution of structured language model programs. *Advances in neural information processing systems*, 37:62557–62583, 2024.

Siqi Zhu, Xuyan Ye, Hongyu Lu, Weiye Shi, and Ge Liu. The many faces of on-policy distillation: Pitfalls, mechanisms, and fixes. *arXiv preprint arXiv:2605.11182*, 2026.

Zilin Zhu, Chengxing Xie, Xin Lv, and slime Contributors. slime: An llm post-training framework for rl scaling. https://github.com/THUDM/slime, 2025. GitHub repository. Corresponding author: Xin Lv.

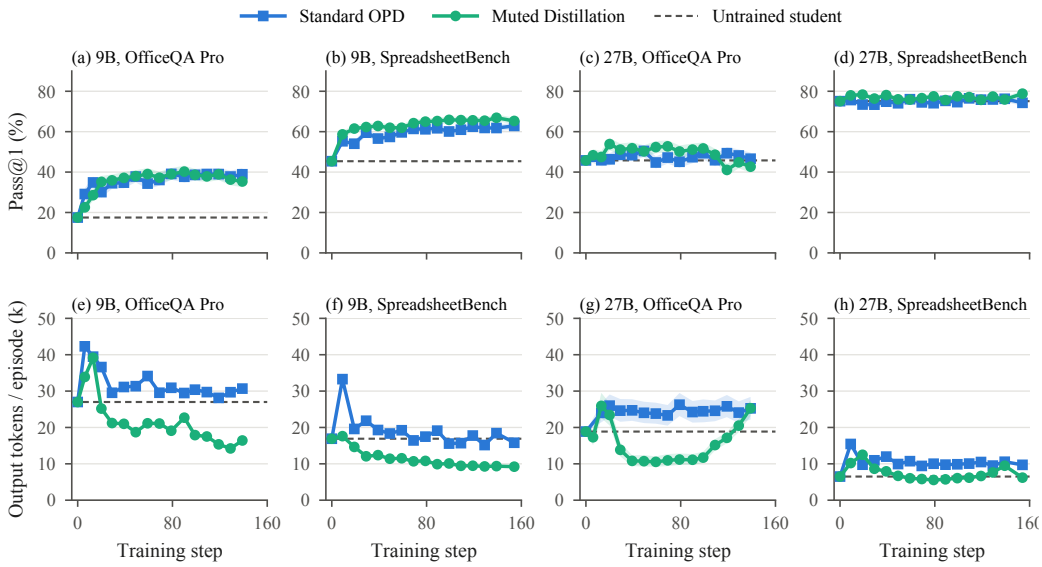


Figure 8: Offline evaluation of every saved checkpoint of the Qwen3.5-9B and Qwen3.5-27B students trained with the Qwen3.5-397B-A17B teacher, under the protocol of Figure 5. Shaded bands are 95% confidence intervals where the evaluation recorded them; the dashed line marks the untrained student.

A Training Curves for Larger Students

Figure 8 shows the per-checkpoint curves behind the 9B and 27B points of Figure 1. On SpreadsheetBench the picture matches the 4B student: Muted Distillation is at least as accurate as standard OPD throughout training and ends well below it in tokens (9.2k vs. 15.8k at 9B, 6.2k vs. 9.7k at 27B), with higher final pass@1 (65.3 vs. 62.8 and 78.8 vs. 74.3). On OfficeQA Pro the token saving is again large at 9B (16.4k vs. 30.7k at the final checkpoint) but the final accuracy is lower (35.3 vs. 38.9), and at 27B the two recipes converge: the Muted Distillation student peaks early (53.9 pass@1 at step 20, the point used in Figure 1) and then declines, ending at 42.7 pass@1 and 25.3k tokens against 46.6 and 25.3k for OPD. The larger students are therefore where Muted Distillation is least reliable on OfficeQA Pro, and the best-checkpoint comparison in Figure 1 should be read together with these curves.

B Best and Final Checkpoints

The main text reports final checkpoints in its tables and best checkpoints in the Pareto plot. Table 3 lists both so that the two can be reconciled. For the 4B student and for every SpreadsheetBench run the choice makes little difference: best and final differ by at most 2.7 points of pass@1 and the Muted Distillation student is ahead of standard OPD under either reading, with the token gap essentially unchanged. The exception is OfficeQA Pro at 9B and 27B, where the Muted Distillation runs peak early (steps 90 and 20) and decline afterwards, so the best-checkpoint comparison in Figure 1 is more favorable to Muted Distillation than the final-checkpoint one in Figure 8. We therefore state the size-transfer claim of Section 4 in terms of the Pareto frontier, which uses the best checkpoint of every recipe including standard OPD, and note the final-checkpoint reversal explicitly in Appendix A.

C Full Ablation

Table 4 extends Table 1 with the remaining variants we trained. Three observations support the reading in Section 3. First, every variant in which the teacher reads the student's thought leaves the student above its untrained length (24.4k and 14.8k tokens) on at least one benchmark, including

Table 3: Best and final checkpoints of every standard OPD and Muted Distillation run with the Qwen3.5-397B-A17B teacher. *Best* is the evaluated checkpoint with the highest pass@1 (the points of Figure 1); *final* is the last checkpoint of the run (the rows of Tables 1 and 2 and the endpoints of Figures 5 and 8). Tokens are mean output tokens per episode in thousands. Three sweeps are incomplete (27B OPD on OfficeQA Pro, 12 of 15 checkpoints; 9B OPD on SpreadsheetBench, 14 of 15; 27B Muted Distillation on SpreadsheetBench, 10 of 15), so their best checkpoint is the best among those evaluated.

		Best checkpoint			Final checkpoint		
Student	Recipe	Step	Pass@1	Tokens	Step	Pass@1	Tokens
OfficeQA Pro							
4B	Standard OPD	59	35.2	33.9	139	35.0	36.0
4B	Muted Distillation	119	39.1	12.5	139	37.7	15.6
9B	Standard OPD	79	39.1	30.9	139	38.9	30.7
9B	Muted Distillation	90	40.2	22.7	139	35.3	16.4
27B	Standard OPD	49	50.5	24.0	139	46.6	25.3
27B	Muted Distillation	20	53.9	23.4	139	42.7	25.3
SpreadsheetBench							
4B	Standard OPD	139	57.4	22.2	154	55.5	19.5
4B	Muted Distillation	129	61.3	9.3	154	60.3	9.1
9B	Standard OPD	154	62.8	15.8	154	62.8	15.8
9B	Muted Distillation	139	66.9	9.4	154	65.3	9.2
27B	Standard OPD	109	76.5	10.1	154	74.3	9.7
27B	Muted Distillation	154	78.8	6.2	154	78.8	6.2

Table 4: All trained variants of the Qwen3.5-4B student at the final checkpoint (OfficeQA Pro: step 139; SpreadsheetBench: step 154), same protocol as Table 1. *Teacher sees* is what the teacher conditions on in place of the student's thought (the thought itself, an empty block, or a thought the teacher writes itself); *student keeps* is whether the student's own sampled thought stays in its context; *loss on* is the set of trained tokens. First-t trains the first 130 (OfficeQA Pro) or 280 (SpreadsheetBench) response tokens.

				OfficeQA Pro		SpreadsheetBench	
Method	Teacher sees	Student keeps	Loss on	Pass@1	Tokens	Pass@1	Tokens
Standard OPD	thought	yes	response	35.0	36.0	55.5	19.5
Think-only	thought	yes	thought	30.9	27.4	57.9	16.7
Mask-think	thought	yes	command	28.8	31.9	55.2	19.3
Mask-think-tail	thought	yes	tail	32.0	26.8	55.5	18.3
First-t	thought	yes	first t	32.1	36.0	56.7	20.6
Teacher-think	own thought	yes	tail	25.1	39.3	49.8	16.3
AOD	empty	no	command	32.4	21.0	52.1	15.7
AOD on-policy	empty	yes	command	33.6	14.2	50.3	12.8
Muted Distillation	empty	yes	tail	**37.7**	**15.6**	**60.3**	**9.1**

first-t, which trains only the opening of each turn. Second, letting the teacher write its own thought before scoring the tail (teacher-think) is the worst variant on both benchmarks: the teacher's own reasoning is no better a context for judging the student's actions than the student's reasoning, and it is far more expensive to obtain. Third, among the muted variants the loss set still matters. Removing the student's thought from its own context as well (AOD) or training only the command (AOD on-policy) both lose accuracy relative to Muted Distillation, which keeps the student's thought and trains the whole visible tail, prose included.

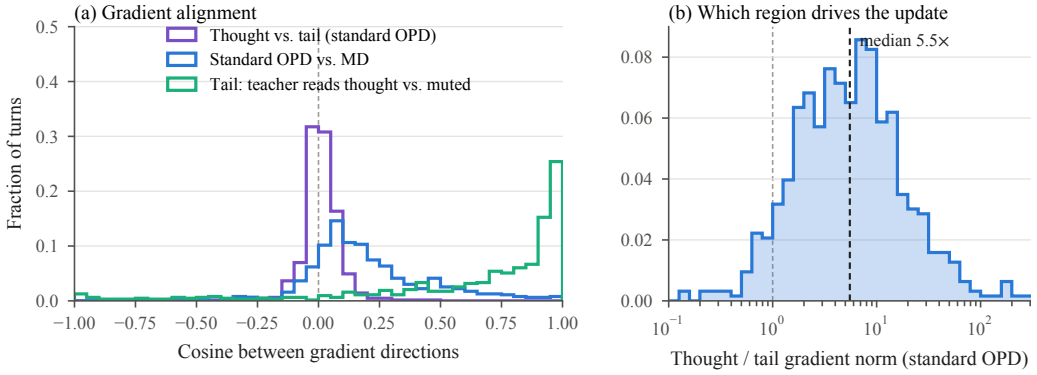


Figure 9: Gradients of the OPD loss on the untrained Qwen3.5-4B student, computed on 630 saved OfficeQA Pro training turns (32 prompts) that contain both a thought and a tail, with the Qwen3.5-397B-A17B teacher. Each turn's gradient is taken over all language-model parameters for the local, unclipped loss of Eq. 1 restricted to one token region and one teacher context. (a) Cosine between the thought-region and tail-region gradients of standard OPD; between the full standard-OPD gradient and the Muted Distillation gradient (Eq. 3); and between the tail-region gradient with the teacher reading the thought and with it hidden. (b) Ratio of the thought-region to the tail-region gradient norm under standard OPD.

Table 5: The same quantities at the final trained checkpoints (OfficeQA Pro: step 139; SpreadsheetBench: step 154) of the standard OPD and Muted Distillation students. Each entry is the mean over four fixed four-task batches of saved turns (16 turns per checkpoint), with the range across batches in brackets. The thought is scored with the teacher reading the full turn; "MD direction" is the gradient of Eq. 3 on the same turn.

Benchmark	Checkpoint	Thought vs. tail cosine	Standard OPD vs. MD direction
OfficeQA Pro	Standard OPD	0.016 [−0.010, 0.038]	0.157 [0.110, 0.216]
OfficeQA Pro	Muted Distillation	0.008 [−0.012, 0.018]	0.115 [0.029, 0.269]
SpreadsheetBench	Standard OPD	0.011 [−0.018, 0.024]	0.208 [0.023, 0.420]
SpreadsheetBench	Muted Distillation	−0.044 [−0.094, 0.002]	0.121 [−0.114, 0.483]

D Gradient Analysis

Section 3 argues from per-token KL values that the teacher's signal on the thought is weak and that it dilutes the signal on the tail. Gradients let us ask the question directly: how much of a standard OPD update is spent on the thought, and does that part help or hurt the tail? We replay saved turns through the untrained 4B student, recompute the student's log-probabilities with autograd, and take the gradient of the loss of Eq. 1 with respect to all language-model parameters, separately for the thought tokens, for the tail tokens with the teacher reading the thought, and for the tail tokens with the thought hidden (the Muted Distillation gradient of Eq. 3). No parameters are updated, so all views share the same student, the same sampled tokens, and the same turns; only the loss mask and the teacher's context differ. Figure 9 summarizes the 630 OfficeQA Pro training turns, and Table 5 repeats the measurement at the final trained checkpoints.

Thought and tail gradients are orthogonal. The cosine between the thought-region and tail-region gradients of standard OPD is centered on zero (median 0.007, interquartile range -0.03 to 0.05); 86% of turns have $|\cos| < 0.1$ and 45% are negative. Supervising the thought therefore moves the student in a direction that is essentially unrelated to the direction that improves its actions. It neither reinforces nor systematically opposes the tail update; it is a separate objective sharing the same parameters. The same holds at the trained checkpoints of both recipes on both benchmarks (Table 5, all means within ± 0.05).

Table 6: Token savings conditioned on success, and the length tail, at the final checkpoints of Table 1 and Appendix A. *Successful attempts* averages output tokens over successful attempts only, on tasks that both recipes solve at least once, weighting tasks equally; *always solved* restricts further to tasks that both recipes solve in all eight attempts. *At limit* is the share of episodes that end by exhausting the step cap or the context window; P90 is the 90th percentile of output tokens per episode in thousands.

	Successful attempts (tokens, k)				Always solved	At limit (%)		P90 tokens (k)	
Benchmark, student	Tasks	OPD	MD	Reduction (%) [CI]	Red. (%) (tasks)	OPD	MD	OPD	MD
OfficeQA Pro, 4B	73	23.7	11.3	52.3 [45.6, 58.3]	39.7 (9)	12.6	1.3	75.1	32.1
SpreadsheetBench, 4B	307	15.3	7.5	51.1 [47.5, 54.4]	36.4 (95)	7.4	3.0	53.0	20.8
OfficeQA Pro, 9B	71	22.9	11.6	49.4 [42.9, 55.0]	36.0 (12)	8.7	0.9	64.3	36.3

But the thought dominates the update. The thought holds 79% of the response tokens in these turns, and its gradient is larger than the tail's in 93% of turns, by a median factor of 5.5 (Figure 9b). Consequently the full standard-OPD gradient points mostly along the thought direction: its cosine with the Muted Distillation gradient is only 0.15 at the median (0.05 to 0.32 interquartile), and 0.12 to 0.21 on average at the trained checkpoints. Under the batch token-mean normalizer of Section 4.2, standard OPD thus spends most of every step on an orthogonal objective whose per-token signal is the weak, padding-tolerant one of Figure 4. This is the gradient-level form of the dilution argument: the tail signal is not cancelled, it is outweighed.

Muting changes the tail gradient little, and slightly strengthens it. The gradient on the tail tokens with the thought hidden from the teacher is well aligned with the gradient on the same tokens with the teacher reading the thought (median cosine 0.83; above 0.5 in 75% of turns), with a slightly larger norm (median ratio 1.16). Hiding the thought therefore does not replace the tail signal with a different one; it keeps the direction and removes the thought gradient that outweighed it, consistent with the larger per-token KL on tail tokens under the muted view in Figure 4a. The turns where the two tail gradients disagree (cosine below zero in 10% of turns) are the cases in which the teacher's judgment of an action depends on the student's reasoning, which is the behavior Muted Distillation deliberately forgoes.

These are local, unclipped gradients of single turns at fixed parameters, on 32 correlated prompts, and they do not reconstruct the historical PPO minibatches or Adam-preconditioned updates; the trained-checkpoint pilot uses 16 turns per checkpoint. They describe where the signal goes, not what the optimizer makes of it over training; the training comparisons of Table 4 are the test of the consequence.

E Where the Token Savings Come From

A shorter mean episode could come from failed episodes that give up sooner rather than from more economical solutions. To separate the two, we compare the saved evaluation episodes of the final standard OPD and Muted Distillation checkpoints (8 attempts per task, temperature 1.0; 4B on both benchmarks and 9B on OfficeQA Pro, 12,784 episodes in total) and condition on the outcome. Confidence intervals are 95% paired task-bootstrap intervals over the evaluation tasks, keeping the eight attempts of a task together; they describe variation across tasks for these trained checkpoints, not variation across training seeds.

The savings survive conditioning on success. Table 6 (left) restricts the comparison to tasks that both recipes solve at least once and to the successful attempts on those tasks. The Muted Distillation student still uses about half the tokens of the OPD student: 52%, 51%, and 49% fewer on the three comparisons, with confidence intervals that exclude any reduction below 40%. On the stricter set of tasks that both recipes solve in all eight attempts the saving is 36–40%, and on SpreadsheetBench these always-solved tasks take essentially the same number of tool steps under both recipes (11.0 vs. 11.2) while their tokens fall from 6.3k to 4.0k. The savings are therefore not an artifact of cheaper failures, and on the easiest tasks they come from shorter turns rather than from skipping parts of the tool workflow.

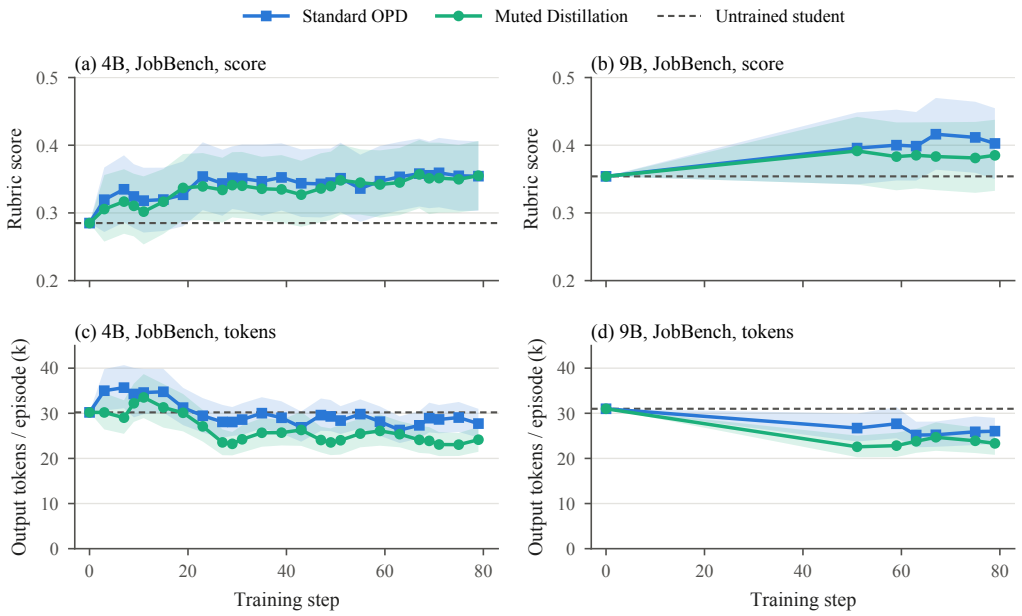


Figure 10: JobBench: offline evaluation of the saved checkpoints of the Qwen3.5-4B and Qwen3.5-9B students trained on the main split with the Qwen3.5-397B-A17B teacher and evaluated on the easy split (8 attempts per task, temperature 1.0, 120-step cap, 95% task-bootstrap intervals). The 9B run was evaluated from step 51 onward, with an 8k-token per-turn budget; the 4B run uses 16k. The dashed line is the untrained student under the same protocol.

Muting removes the long tail. Mean tokens understate the difference at the tail, which is what determines worst-case latency and context pressure. Table 6 (right) shows that the 90th-percentile episode of the OPD student is 2–2.5× longer than that of the Muted Distillation student (75.1k vs. 32.1k tokens on OfficeQA Pro at 4B), and that the share of episodes ending by exhausting the step cap or the context window falls from 12.6% to 1.3% on OfficeQA Pro and from 7.4% to 3.0% on SpreadsheetBench at 4B. Much of the OPD tail is unfinished thinking: only 2.1% of its OfficeQA Pro turns lack a closing `</think>`, but those turns average about 8.1k tokens each and account for 21.4% of all output, against 1.2% of output for Muted Distillation. In the most extreme saved episode the OPD student emits 244k tokens across 26 turns that each hit the 8,192-token generation limit, revisiting the same table interpretation without producing a usable action, before the context window is exhausted; the Muted Distillation attempt on the same task is also incorrect but submits after 101k tokens. Such cases are cheaper failures rather than extra successes, which is why the success-conditioned comparison above is the one that supports the efficiency claim.

F A Third Benchmark: JobBench

OfficeQA Pro and SpreadsheetBench both have verifiable rewards. To check that the behavior of Muted Distillation does not depend on that, we also train on JobBench, a benchmark of 128 professional-deliverable tasks in which the agent receives a folder of source documents (spreadsheets, documents, PDFs, and databases; about ten files per task) and a brief from one of 35 white-collar occupations, and must write its deliverables into an output directory. The reward is an LLM judge: each task comes with a weighted rubric, each rubric item passes only if all of its sub-criteria pass, and the score is the normalized weighted sum. We train on the 65-task main split and evaluate on the 63-task easy split, whose scenarios are unrelated and share only the occupation labels, using the same harness as OfficeQA Pro and the same training recipe as Section 4.2. Grading uses a different judge from the public leaderboard (Gemini 3.5 Flash at low reasoning effort), so absolute scores are not comparable to published numbers; the within-figure comparisons are like for like.

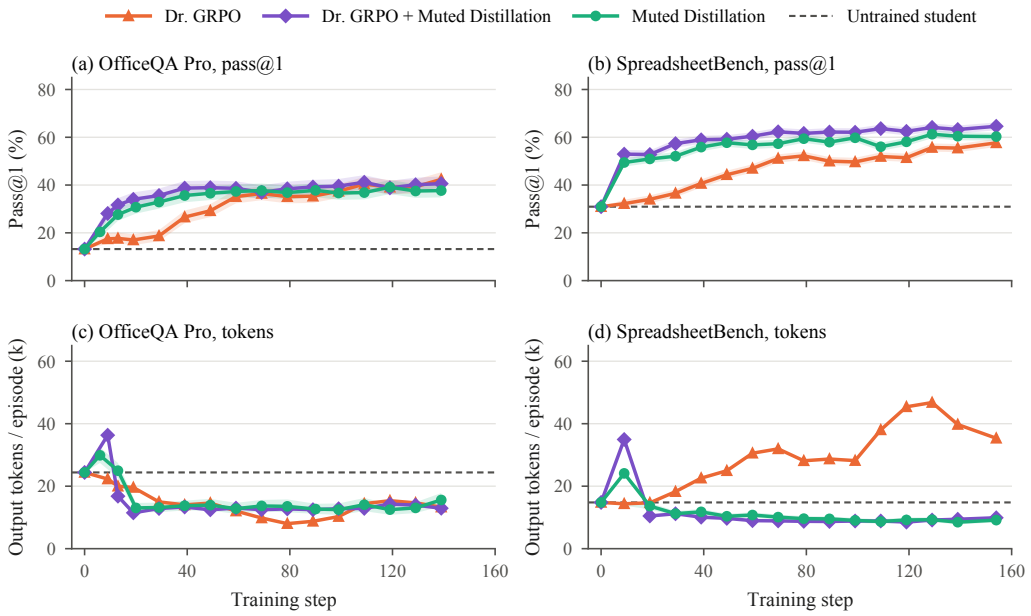


Figure 11: Offline evaluation of every saved checkpoint of the Qwen3.5-4B student trained with the task reward alone (Dr. GRPO), with the task reward plus the Muted Distillation signal, and with Muted Distillation alone, under the protocol of Figure 5.

Table 7: Final checkpoints of the three recipes in Figure 11 (step 139 on OfficeQA Pro, 154 on SpreadsheetBench), and their transfer to OfficeQA Pro v2 and SpreadsheetBench 2 under the protocol of Table 2. Dr. GRPO runs use 8 episodes per prompt; Muted Distillation alone uses one.

	OfficeQA Pro		SpreadsheetBench		OfficeQA Pro v2		SpreadsheetBench 2	
Method	Pass@1	Tokens	Pass@1	Tokens	Pass@1	Tokens	Score	Tokens
Dr. GRPO	**42.4**	**13.3**	57.7	35.3	**15.6**	**16.2**	16.6	79.8
Dr. GRPO + Muted Distillation	40.6	12.9	**64.6**	9.9	9.6	31.9	**38.5**	**25.1**
Muted Distillation	37.7	15.6	60.3	**9.1**	15.1	31.1	37.6	26.0

Figure 10 shows the curves. At 4B the two recipes reach the same score (0.355 for Muted Distillation vs. 0.354 for standard OPD at the final checkpoint, step 79, from 0.285 untrained), while the Muted Distillation student uses 13% fewer output tokens (24.1k vs. 27.7k) and sits below the OPD student in tokens at every evaluated checkpoint. The saving is smaller than on the two main benchmarks, where standard OPD inflates episodes well above the untrained length; on JobBench both recipes end below it (30.2k untrained), so there is less inflation for muting to remove. For reference, the online SFT baseline at the same step scores 0.342 at 37.7k tokens. At 9B the picture is mixed: Muted Distillation is again cheaper (23.3k vs. 26.0k) but scores lower at the final checkpoint (0.385 vs. 0.403, both well above the 0.354 of the untrained student) with overlapping confidence intervals, mirroring the 9B OfficeQA Pro result of Appendix A. We read JobBench as evidence that the token saving of muting does not depend on a verifiable reward, and as a second reminder that it does not guarantee an accuracy gain for the larger student.

G Combining Muted Distillation with Reinforcement Learning

Muted Distillation needs no reward, but nothing prevents adding one. We train a variant whose per-token advantage is the sum of the Dr. GRPO advantage (the group-centered task reward with a constant length normalizer, Section 4.2) and the muted distillation signal on the tail, using Dr. GRPO's rollout geometry of 8 episodes per prompt. Figure 11 and Table 7 compare it with each ingredient alone.

The two benchmarks behave differently. On SpreadsheetBench, where the task reward alone drives the student into very long episodes (35.3k tokens at the final checkpoint, 2.4× the untrained student, and up to 46.8k during training), adding the muted signal removes the inflation and improves accuracy at the same time: 64.6 vs. 57.7 pass@1 at 9.9k tokens, which is also 4.3 points above Muted Distillation alone at nearly the same length. The combination is the best SpreadsheetBench student we trained, and its advantage carries over to SpreadsheetBench 2 (38.5 target-cell accuracy at 25.1k tokens, against 16.6 at 79.8k for Dr. GRPO). On OfficeQA Pro the task reward alone is already short and accurate (42.4 pass@1 at 13.3k), and the combination changes little at the final checkpoint (40.6 at 12.9k), apart from a larger early spike in tokens (36.3k at step 9, against 29.9k for Muted Distillation alone and none for Dr. GRPO); its transfer to OfficeQA Pro v2 is also weaker than either ingredient's. We read this as the two signals pulling in different directions when the reward is informative: the verifiable reward already shapes the tail, and the distillation term adds a second target for the same tokens.

Two caveats apply. These are single runs, and the Dr. GRPO recipes use 8× the rollout compute per step of the distillation runs, so the comparison with Muted Distillation alone is not compute-matched. The practical conclusion is narrower than the SpreadsheetBench numbers suggest: when a verifiable reward is available, Muted Distillation can be added to it at no cost in tokens, and it helps most where the reward alone fails to keep episodes short.

H Harness and Prompts

Agent loop. Both benchmarks use the same loop. The system prompt (below) and a task message open the conversation. In each turn the model generates a thought and a response; the harness extracts exactly one fenced `bash` block from the response, executes it in the task's sandbox container, and returns the command's output as the next user message, truncated to 12,000 characters by keeping the first and last 6,000 and inserting `[... omitted N characters ...]`. A response without exactly one fenced block receives a fixed format-error message (*Format error: respond with exactly one fenced command block, for example:* ...) and costs a turn. Commands time out after 30 seconds. The episode ends when the model calls `submit`, when it exhausts the turn cap (120 on OfficeQA Pro, 50 on SpreadsheetBench), or when the context budget is exceeded (64k tokens during training; the model's native 262k context at evaluation), with at most 8,192 generated tokens per turn.

OfficeQA Pro sandbox. A `python:3.11-slim` container with the parsed corpus mounted read-only at `officeqa_corpus/` as one text file per bulletin (`treasury_bulletin_YYYY_MM.txt`) and a writable workspace containing `question.txt`. The container has no network access. The task message is:

```
Question: <question text>

Workspace: /workspace
Useful files: question.txt, source_files.txt, manifest.json.
Corpus path: officeqa_corpus/

Search and read the parsed Treasury Bulletin corpus, calculate as needed
    , then submit the final answer.
```

The system prompt, with the corpus-specific fragments rendered for the Treasury Bulletin corpus, is:

```
You are an autonomous OfficeQA agent and an expert in answering
    questions related to the U.S. Treasury and economy.

The Treasury Bulletin corpus is available in the workspace at `
    officeqa_corpus/`. Use filesystem search to locate relevant
    documents and sections; these documents are long, so do not read the
     whole corpus or whole files unless needed.
```

```
If multiple documents report the same metric, use the most up-to-date
    revision unless the question specifies an exact date or document. Do
     not rely on the first matching value you encounter.

You can issue exactly one command per turn. Put the command in exactly
    one fenced bash block and do not write anything after the block.

Available actions:
- Run shell commands such as `ls`, `find`, `grep`/`rg`, `sed`, `head`, `
    cat`, and `python3` to search, read evidence, and calculate.
- `submit "<answer>"`: Submit the final answer and finish the task.

Useful corpus patterns:
- Filenames look like `treasury_bulletin_YYYY_MM.txt`.
- Prefer targeted searches such as `grep -Rni -A 3 -B 3 "phrase"
    officeqa_corpus/treasury_bulletin_1954_*.txt`.
- Use Python for arithmetic, statistics, regressions, and careful unit
    conversion.

REQUIRED ANSWER FORMAT:
When you have the final answer, call `submit "<answer>"` with only the
    answer value inside the quotes.

Example:
```bash
grep -Rni -A 2 -B 2 "national defense" officeqa_corpus/
    treasury_bulletin_1941_01.txt | head -50
```

Example:
```bash
submit "123.45"
```

Use maximum precision in calculations unless the question specifies
    otherwise. Never ask follow-up questions.
```

SpreadsheetBench sandbox. A container with Python 3.11, `openpyxl`, `pandas`, `numpy`, `xlsxwriter`, `matplotlib`, LibreOffice, and a `view_xlsx` helper, mirroring the benchmark's official image, with no network access. Only the first test case's workbook is mounted; golden workbooks and the other test cases never enter the container. The task message gives the instruction, its type (cell- or sheet-level), the answer position, and the fixed paths of the visible input, the visible output, and the reusable program:

```
## SpreadsheetBench v1 task
Instruction: <instruction text>
Instruction type: <Cell-Level Manipulation | Sheet-Level Manipulation>
Answer position: <e.g. D2:D6>

Visible input: /workspace/<input>.xlsx
Visible output: /workspace/<output>.xlsx
Reusable program: /workspace/solution.py

Create a general solution.py, run it on the visible input, verify the
    output, and submit. The evaluator will run the program on held-out
    workbook variants.
```

The system prompt is:

```
You are an autonomous spreadsheet automation engineer working in a Linux
     sandbox.
```

```
Your job: solve the spreadsheet instruction by writing a reusable Python
     program at /workspace/solution.py. The program must read the input
    path from $SSB_INPUT and save the completed workbook to $SSB_OUTPUT.
     It will be evaluated on additional workbook variants that have the
    same task but different values, so never hard-code values copied
    from this workbook.

You can issue exactly one command per turn. Put the command in exactly
    one fenced bash block and do not write anything after the block.

Available actions:
- Run shell commands. python3 has openpyxl, pandas, numpy, xlsxwriter,
    and matplotlib installed; LibreOffice (`soffice`) is available.
- `view_xlsx <file> list`: list workbook sheets.
- `view_xlsx <file> content "<sheet>" [start_row [end_row]]`: inspect
    values and formulas.
- `submit`: finish. It succeeds only after both /workspace/solution.py
    and $SSB_OUTPUT exist.

Required workflow:
1. Use the supplied Instruction type and Answer position, then inspect
    $SSB_INPUT and infer a general solution from labels, formulas, and
    workbook structure. Answer position identifies the destination cells
    ; it does not contain their correct values.
2. Write /workspace/solution.py. It must use os.environ["SSB_INPUT"] and
     os.environ["SSB_OUTPUT"], preserve unrelated cells and formatting,
    and derive results from workbook contents rather than fixed answers.
3. Run `python3 /workspace/solution.py` to produce $SSB_OUTPUT for the
    visible workbook.
4. Reopen and verify the output, then run `submit`.

Example:
```bash
view_xlsx "$SSB_INPUT" list
```

Example:
```bash
submit
```

Never ask follow-up questions. Golden workbooks and held-out variants
    are not available in the sandbox.
```

Grading. On OfficeQA Pro the submitted string is compared with the reference after normalization (Unicode minus signs, whitespace, punctuation, and unit abbreviations); numerical answers match within a 1% relative tolerance, multi-part and date answers require agreement on both their numbers and their text, and non-numerical answers use fuzzy text matching. On SpreadsheetBench the submitted `solution.py` is re-run on every test case of the instruction in a fresh container, each output workbook is recalculated with LibreOffice, and the cells in the answer position are compared with the golden workbook; an instruction is solved only if every test case matches (hard restriction). The training reward on SpreadsheetBench is the fraction of test cases that match (soft restriction).